2.6 Nucleus

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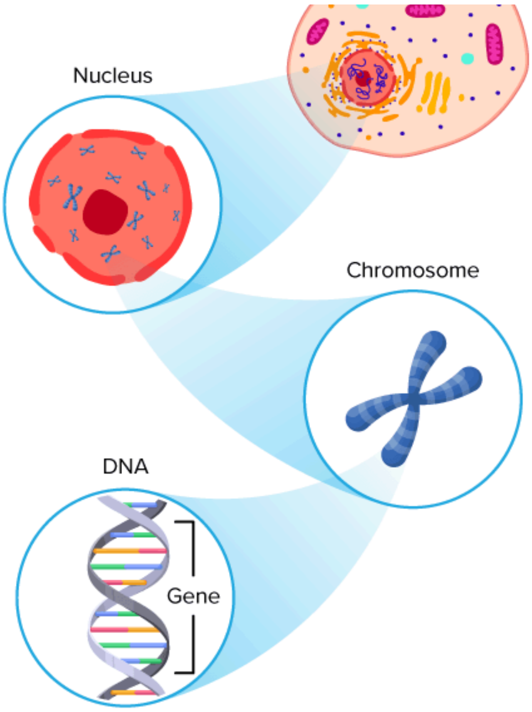
What you will learn

- How DNA is organized inside the nucleus
- The role of the nucleus
- The structure and function of the **nuclear envelope**
- The role of the **nucleolus** in the assembly of **ribosomes**



[Figure 1]

Where does the DNA live?

The answer depends on if the cell is prokaryotic or eukaryotic. The main difference between the two types of cells is the presence of a nucleus. And in eukaryotic cells, DNA lives in the nucleus.

The Nucleus

The **nucleus** is a membrane-enclosed organelle found in most eukaryotic cells. The nucleus is the largest organelle in the cell and contains most of the cell's genetic information (mitochondria also contain DNA, called mitochondrial DNA, but it makes up just a small percentage of the cell's overall DNA content). The genetic information, which contains the information for the structure and function of the organism, is found encoded in DNA in the form of **genes**. A **gene** is a short segment of DNA that contains information to encode an RNA molecule into a protein strand. DNA in the nucleus is organized in long linear strands that are attached to different proteins. These proteins help the DNA coil up for better storage in the nucleus. Think how a string gets tightly coiled up if you twist one end while holding the other end. These long strands of coiled-up DNA and proteins are called **chromosomes.** Each chromosome contains many genes. The function of the nucleus is to maintain the integrity of these genes and to control the activities of the cell by regulating **gene expression**. **Gene expression** is the process by which the information in a gene is "decoded" by various cell molecules to produce a functional gene product, such as a protein molecule.

DID YOU KNOW?

Every cell in the human body has 3 meters of DNA packed inside its nucleus, except gametes which have half the amount of DNA.

The degree of DNA coiling determines whether the chromosome strands are short and thick or long and
Between cell divisions, the DNA in chromosomes is more loosely coiled and forms long, thin strands

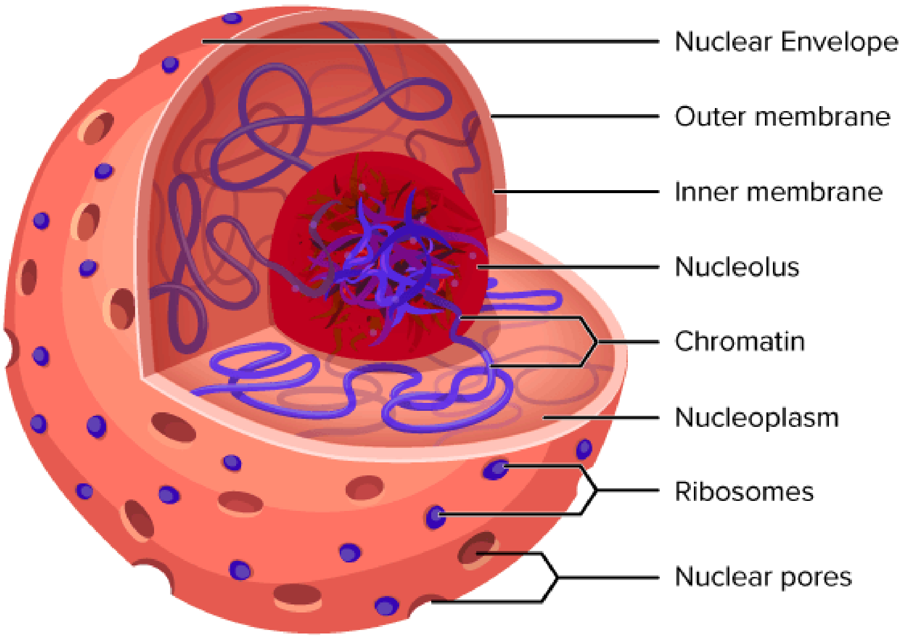
Greek word *chroma* (color), and *soma* (body), due to its ability to be stained strongly by dyes.

The Nuclear Envelope

The **nuclear envelope** is a double membrane of the nucleus that encloses the genetic material. It separates the contents of the nucleus from the cytoplasm. The nuclear envelope is made of two lipid bilayers, an inner membrane and an outer membrane. The outer membrane is continuous with the rough endoplasmic reticulum. Many tiny holes called **nuclear pores** are found in the nuclear envelope. These nuclear pores help to regulate the exchange of materials (such as RNA and proteins) between the nucleus and the cytoplasm.

The Nucleolus

The nucleus of many cells also contains a non-membrane-bound organelle called a **nucleolus**, shown in the **Figure** below. The nucleolus is mainly involved in the assembly of ribosomes. **Ribosomes** are cellular parts made of protein and ribosomal RNA (rRNA), and they build cellular proteins in the cytoplasm. The function of the rRNA is to provide a way of decoding the genetic messages within another type of RNA (called mRNA), into amino acids. After being made in the nucleolus, ribosomes are exported to the cytoplasm, where they direct protein synthesis.



[Figure 2]

The eukaryotic cell nucleus. Visible in this diagram are the ribosome-studded double membranes of the nuclear envelope, the DNA (as chromatin), and the nucleolus. Within the nucleus is a viscous liquid called nucleoplasm, similar to the cytoplasm found outside the nucleus. The chromatin (which is normally invisible), is visible in this figure only to show that it is spread throughout the nucleus.

DID YOU KNOW?

A single mammalian cell may contain up to 10 million ribosomes.

Summary

- The nucleus is a membrane-enclosed organelle, found in most eukaryotic cells, which stores the genetic material (DNA).
- The nucleus is surrounded by a double lipid bilayer, the nuclear envelope, which is embedded with nuclear pores.
- The nucleolus is inside the nucleus and is where ribosomes are made.

Review

1. What is the role of the nucleus of a eukaryotic cell?
2. Describe the nuclear membrane.
3. What are nuclear pores?
4. What is the role of the nucleolus?